Lab 1: Introduction to Packet Tracer, Peer-to-Peer Communication, Study of Cables and its Color Codes

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Aim

To get familiar with Cisco Packet Tracer, set up peer-to-peer communication between two computers through a switch, and study the types of network cables and their color codes.

Tool and Devices Used

- Software: Cisco Packet Tracer
- Devices: 2 PCs (PC-PT) and 1 switch (Switch-PT)
- Cables: 2 copper straight-through Ethernet cables

Addressing Table

Device	Interface	IP Address	Subnet Mask
PC0	FastEthernet0	192.168.1.1	255.255.255.0
PC1	FastEthernet0	192.168.1.2	255.255.255.0

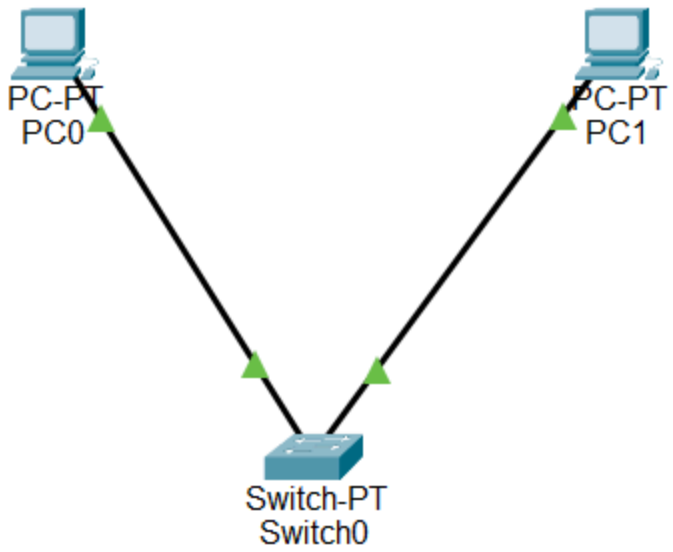
Procedure

1. Launch Cisco Packet Tracer and familiarize yourself with the workspace, device selection and tools.
2. Drag two PCs and one switch onto the workspace.
3. Connect each PC to the switch using copper straight-through Ethernet cables.
4. Click PC0, go to the Desktop tab, open IP Configuration and assign the IP address 192.168.1.1 with the subnet mask 255.255.255.0.

5. Click PC1, go to the Desktop tab, open IP Configuration and assign the IP address 192.168.1.2 with the subnet mask 255.255.255.0.
6. On PC0, open the Command Prompt from the Desktop tab and run `ping 192.168.1.2`.
7. Observe the replies to confirm that the two computers can communicate.
8. Study the cables and their color codes: a straight-through cable is used to connect different devices (for example a computer to a switch), and a crossover cable is used to connect similar devices (for example a computer to a computer).

Output

Figure 1.1: Network topology (PC0 and PC1 connected to Switch0 using straight-through cables)



Result

Two computers were connected through a switch, assigned IP addresses in the same network, and tested using the ping command. The successful replies confirm peer-to-peer communication. The use of straight-through and crossover cables with their color codes was also studied.

Lab 3: Router Configuration (Creating Passwords, Configuring Interfaces)

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Aim

To configure a router in Cisco Packet Tracer by setting the console, enable and VTY passwords, assigning IP addresses to its interfaces, and testing connectivity between two computers through the router.

Tool and Devices Used

- Software: Cisco Packet Tracer
- Devices: 1 Cisco 2911 router and 2 PCs (PC0, PC1)
- Cables: 2 copper straight-through Ethernet cables

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
Router0	GigabitEthernet0/0	192.168.1.1	255.255.255.0	N/A
Router0	GigabitEthernet0/1	192.168.2.1	255.255.255.0	N/A
PC0	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	FastEthernet0	192.168.2.2	255.255.255.0	192.168.2.1

Procedure

1. Open Cisco Packet Tracer on the computer.
2. Drag one 2911 router and two PCs onto the workspace.

3. Connect PC0 to the router interface GigabitEthernet0/0 and PC1 to GigabitEthernet0/1 using copper straight-through cables.
4. Click the router and open the CLI tab. Enter global configuration mode using `enable` and `configure terminal`.
5. Set the console password using `line console 0`, `password cisco` and `login`.
6. Set the enable password using `enable password cisco`.
7. Set the VTY password using `line vty 0 4`, `password cisco` and `login`.
8. Configure the first interface GigabitEthernet0/0 with the IP address 192.168.1.1 and subnet mask 255.255.255.0, and enable it using `no shutdown`.
9. Configure the second interface GigabitEthernet0/1 with the IP address 192.168.2.1 and subnet mask 255.255.255.0, and enable it using `no shutdown`.
10. Open Desktop, then IP Configuration on each PC and assign the IP address, subnet mask and default gateway given in the addressing table.
11. Use the `ping` command from the Command Prompt of the PCs to test connectivity through the router.
12. Log in to the router CLI again to verify that the console and enable passwords are requested, and run `show ip interface brief` to verify the interfaces.

Router Configuration Commands

```
Router> enable
Router# configure terminal
Router(config)# line console 0
Router(config-line)# password cisco
Router(config-line)# login
Router(config-line)# exit
Router(config)# enable password cisco
Router(config)# line vty 0 4
Router(config-line)# password cisco
Router(config-line)# login
Router(config-line)# exit
Router(config)# interface gig0/0
Router(config-if)# ip address 192.168.1.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface gig0/1
Router(config-if)# ip address 192.168.2.1 255.255.255.0
```

```
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# exit
Router# write memory
```

Output

Figure 1: Network topology (PC0 and PC1 connected to Router0 using copper straight-through cables)

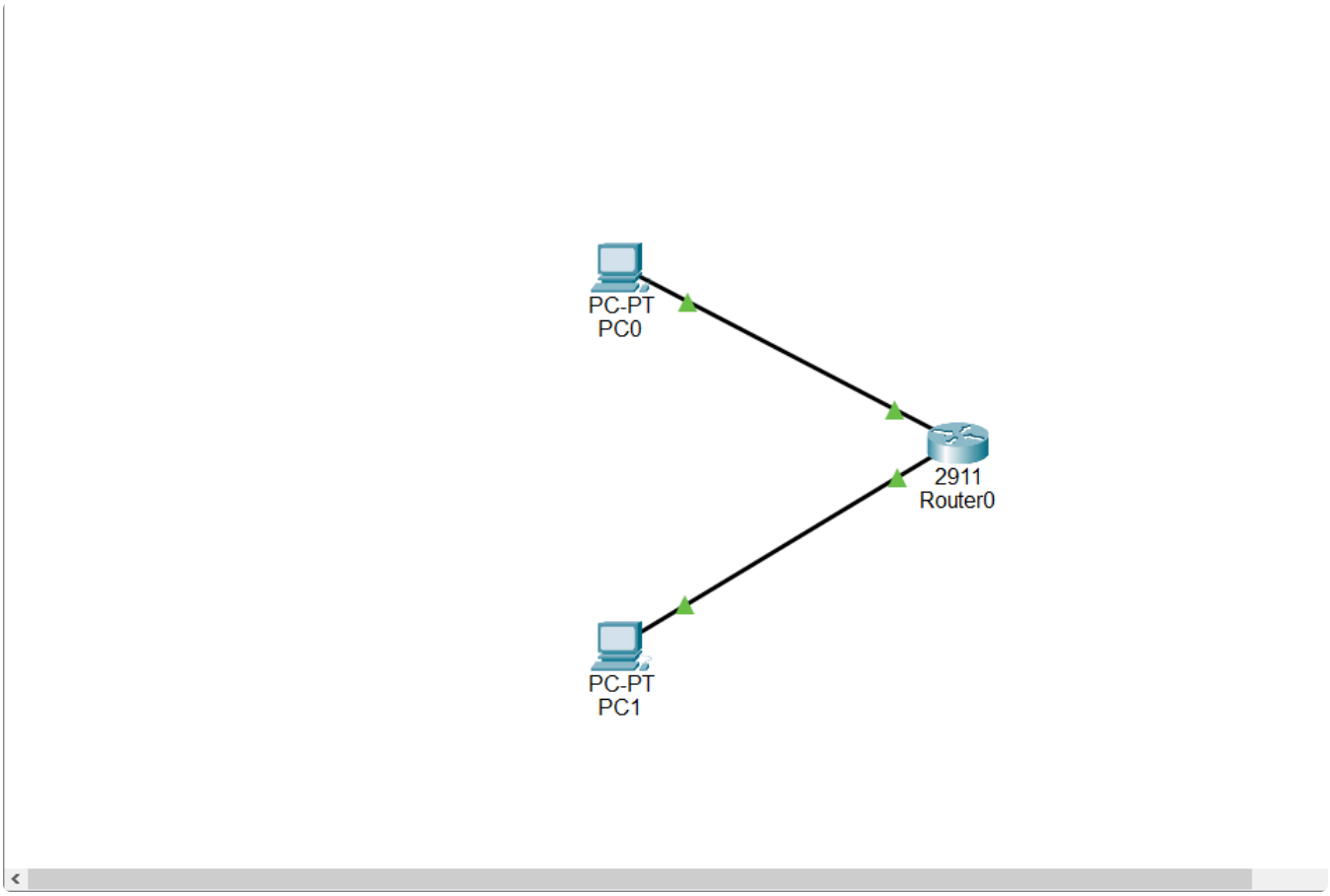


Figure 2: Router CLI showing the password prompts (console and enable) and the `show ip interface brief` output

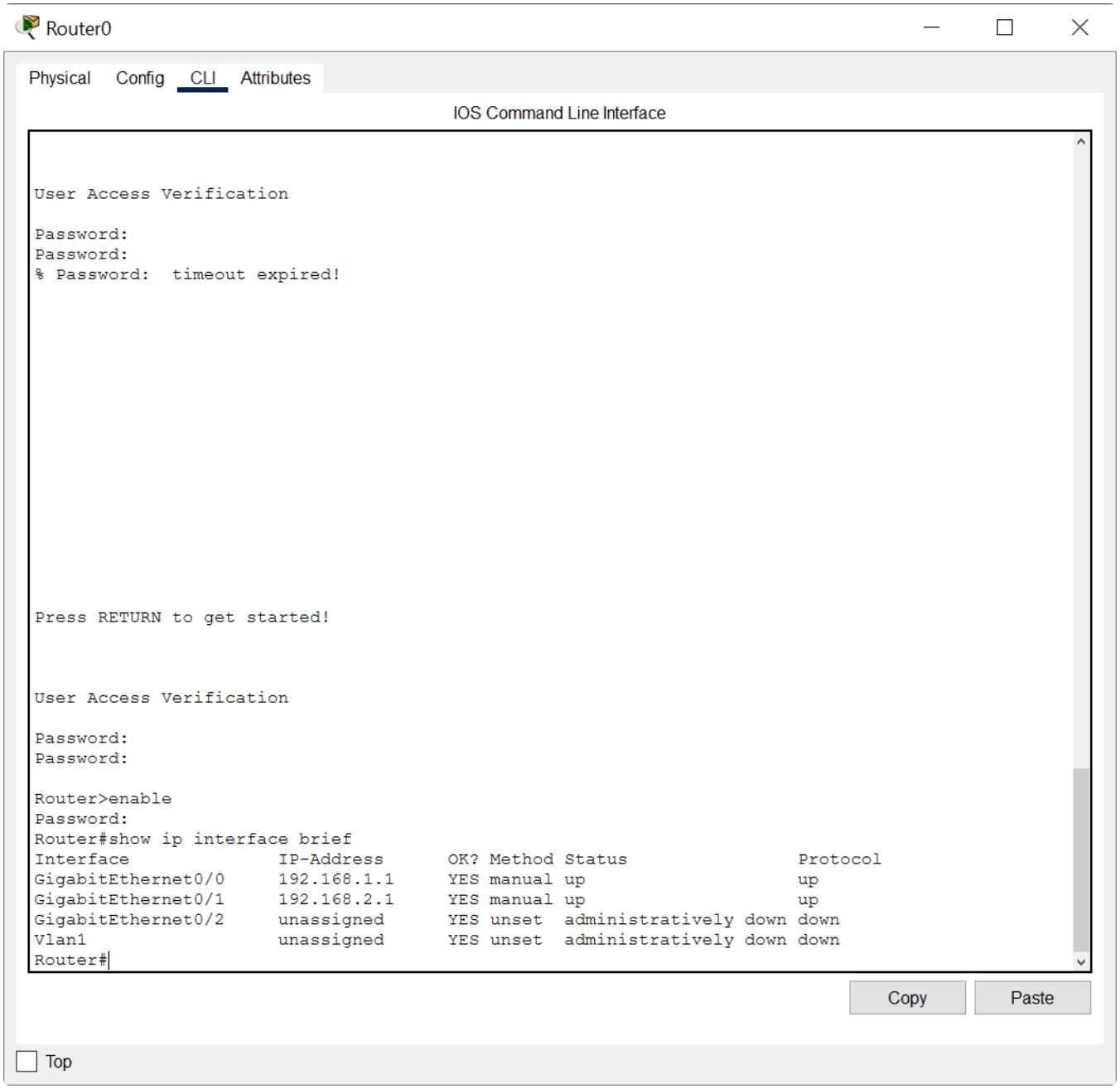


Figure 3: Ping from PC1 (192.168.2.2) to PC0 (192.168.1.2) through the router

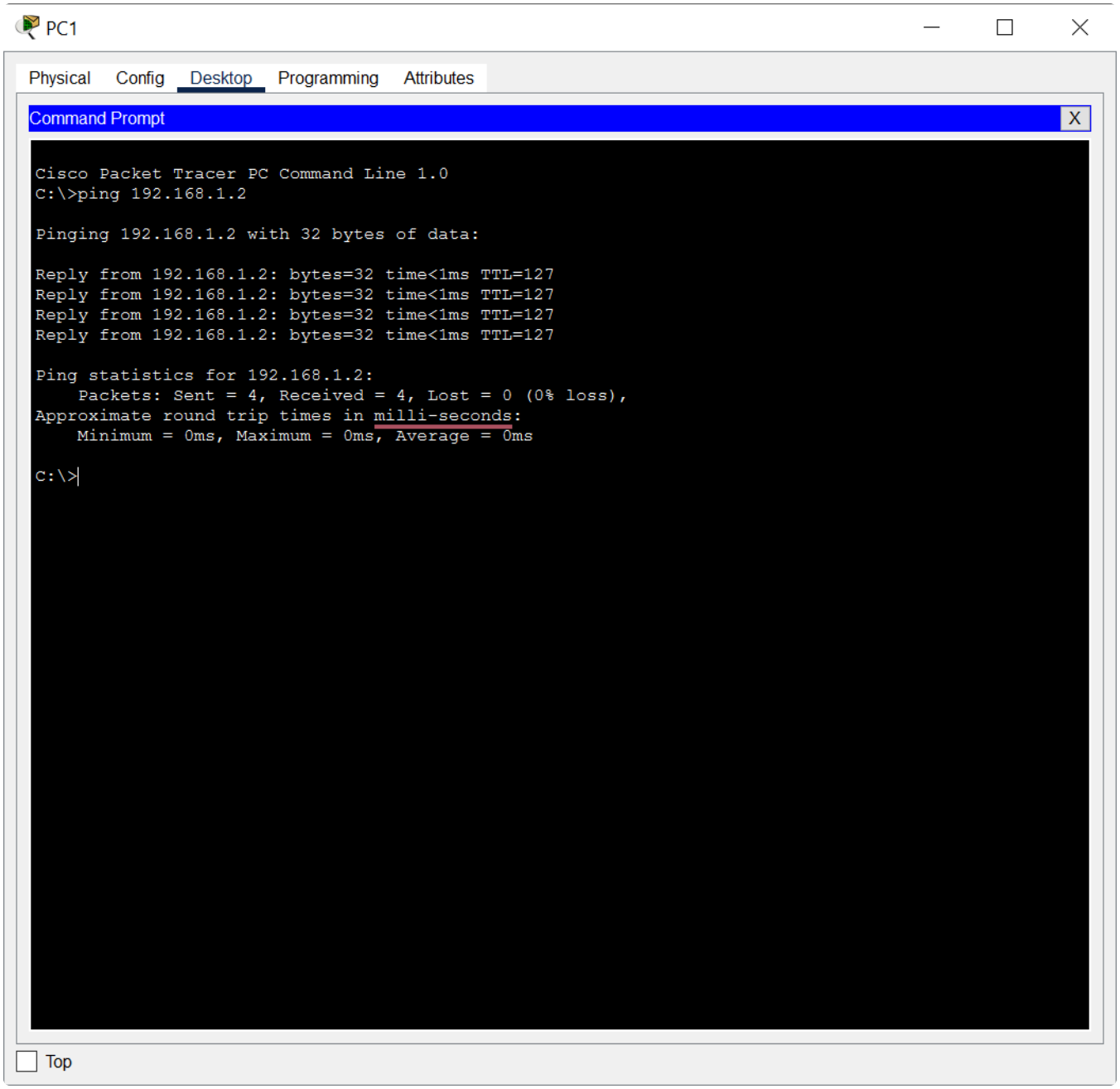
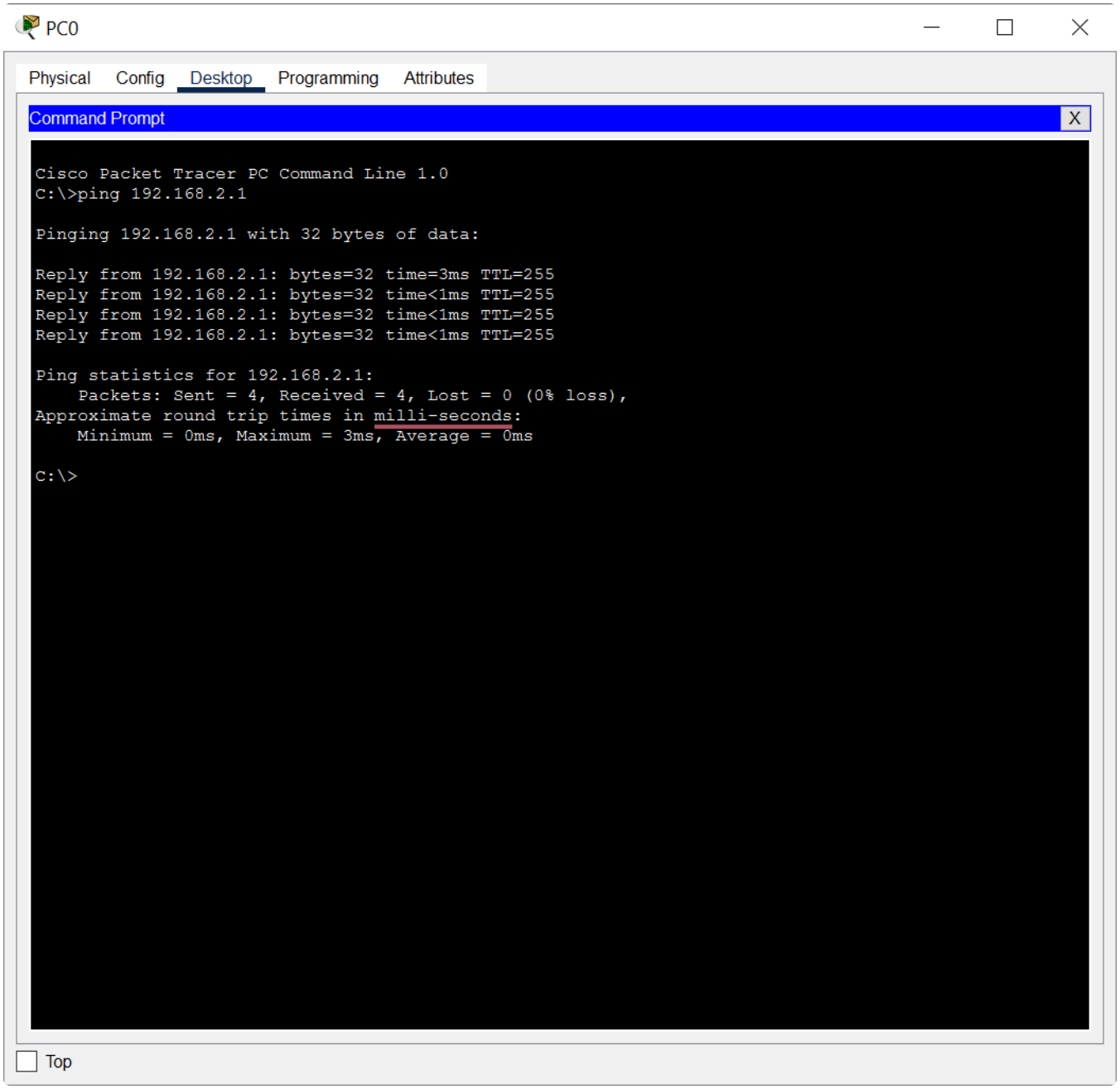


Figure 4: Ping from PC0 (192.168.1.2) to the router interface 192.168.2.1



Result

The router was configured with the console, enable and VTY passwords. The interfaces GigabitEthernet0/0 (192.168.1.1) and GigabitEthernet0/1 (192.168.2.1) were assigned IP addresses and brought up. The `show ip interface brief` output shows both interfaces in the up/up state, and the router asked for the console and enable passwords on login. The ping between the two computers through the router was successful with 0% packet loss, which confirms that the router configuration is correct.